

HAT-P-36b

R-0955

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230216 · created 2026-07-29T15:09:41+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459991.8832 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1212 +/- 0.0098
Transit depth (Rp/Rs)^2	1.47 +/- 0.24 [%]
Orbital Inclination (inc)	89.73 +/- 1.79
Airmass coefficient 1 (a1)	1.0374 +/- 0.0095
Airmass coefficient 2 (a2)	-0.033 +/- 0.018
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#2 - [119, 28]
Optimal Aperture	4.56
Optimal Annulus	14.95
Transit Duration (day)	0.0968 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1212 $\pm$ 0.0098 vs 0.126 (deviation 0.39 σ)
Duration fit vs published	0.0968 d vs 0.0925 d (deviation 2.48 σ)
Mid-time O-C	-0.5 min
Depth SNR	10.0
Residual scatter	0.92 %

Light curve

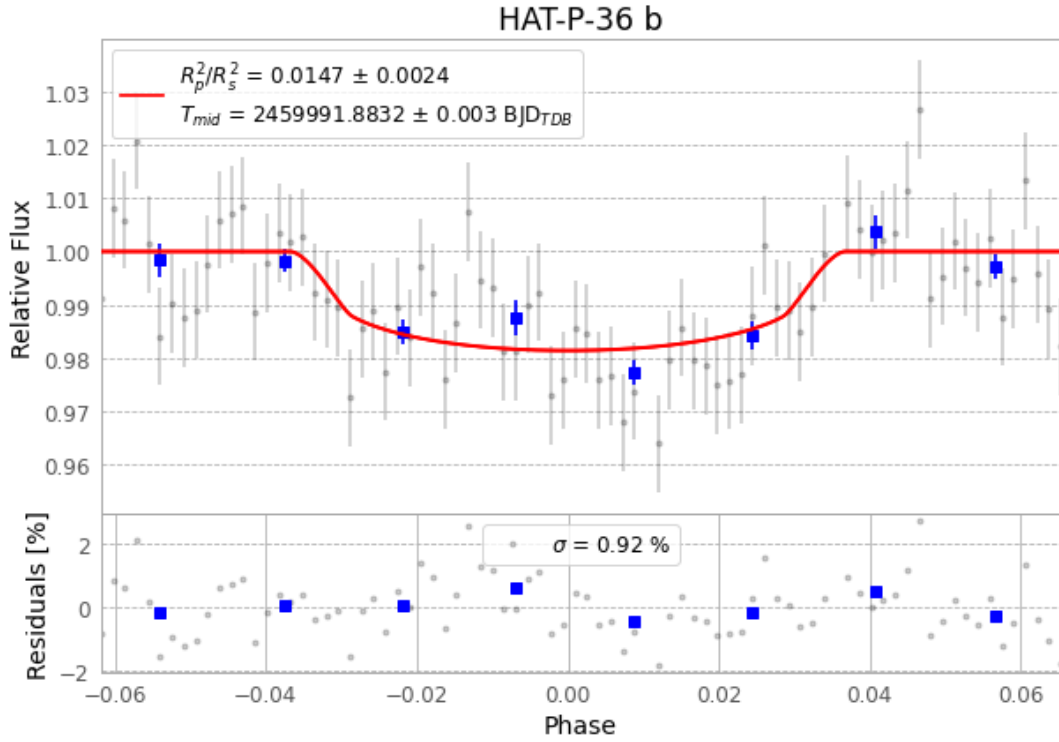


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-02-16.png (SHA-256 e17d1ae3776f0c3a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [578, 113], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 00827fa929865c78...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-36230216070616.FITS → HATP-36230216110915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230216.log (16fec5e1023e...), FinalLightCurve_HAT-P-36 b_2023-02-16.png (e17d1ae3776f...), FinalLightCurve_HAT-P-36 b_2023-02-16.csv (dae29703e547...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 15:09Z.